

$$9 \left[\begin{array}{cccc|c} -21 & -25 & -25 & -23 & \text{Ans} \\ -36 & -40 & -40 & -44 & \\ -18 & -23 & -32 & -37 & \end{array} \right]$$

1.8-Matrix Transformations

Basics:

$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} \rightarrow \text{ordered Pairs } (x, y)$$

$$\text{or } \begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \text{Column vector}$$

In 3D:-

$$\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R} \rightarrow \text{ordered pairs } (x, y, z)$$

$$\text{or } \begin{bmatrix} x \\ y \\ z \end{bmatrix} \rightarrow \text{Column vector}$$

For n^{th} time:

$$\mathbb{R}^n = \mathbb{R} \times \mathbb{R}, \dots, \mathbb{R}^n \rightarrow \text{ordered } n\text{-tuple } (x_1, x_2, x_3, \dots, x_n)$$

$$\text{or } \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Standard Base vectors:-

Building Block of vectors in any coordinate system

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} \rightarrow \text{Standard Basis Vector in } \mathbb{R}^3$$

These are called SBV coz all other vectors made up from these.

Definition : 1

If f is a function with domain $\mathbb{R}^n$ to and codomain $\mathbb{R}^m$ then we say f is transformation.

$$\begin{array}{ccc} f: \mathbb{R}^n & \rightarrow & \mathbb{R}^m \\ \downarrow & & \searrow \\ \text{Domain/Input} & & \text{CoDomain/output} \end{array}$$

where $n=m$ is called operator

Notations:-

$$\begin{array}{ccccccc} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = w_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = w_2 \\ \vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n = w_m \end{array}$$

In Metrics Form.

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_m \end{bmatrix}$$

$A \qquad \qquad \qquad \begin{matrix} x \\ \mathbb{R}^n \end{matrix} \qquad = \qquad \begin{matrix} w \\ \mathbb{R}^m \end{matrix}$

$$\begin{array}{ccc} A & X & = & W \\ \downarrow & \downarrow & & \downarrow \\ \text{input} & & & \text{output} \end{array}$$

$$\begin{array}{ccc} A \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} & = & \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_m \end{bmatrix} \\ \downarrow & & \downarrow \\ \text{input} & & \text{output} \end{array}$$

$$\begin{array}{l} T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m \\ T_A(X) = AX \end{array}$$

Example:

$$\begin{bmatrix} 1 \\ 5 \\ 6 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 5 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 6 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$\downarrow$ $\downarrow$ $\downarrow$
 e_1 e_2 e_3

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 e_1 + x_2 e_2 + x_3 e_3$$

Here $\begin{bmatrix} 1 \\ 5 \\ 6 \end{bmatrix}$ made up from SBV in $\mathbb{R}^3$

Example : 1

Find standard matrix for operator $T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ defined by

$$2x_1 - 3x_2 + x_3 - 5x_4 = w_1$$

$$4x_1 + x_2 - 2x_3 + x_4 = w_2$$

$$5x_1 - x_2 + 4x_3 = w_3 \quad \text{and then compute}$$

$T(1, -3, 0, 2)$ by directly substituting in eqs and then by matrix multiplication.

$$\text{SBV} = \begin{bmatrix} 2 & -3 & 1 & -5 \\ 4 & 1 & -2 & 1 \\ 5 & -1 & 4 & 0 \end{bmatrix}$$

Compute $T(1, -3, 0, 2)$ by Direct subs:

$$w_1 = 2(1) - 3(-3) + 0 - 5(2) = 2 + 9 + 0 - 10 = 1$$

$$w_2 = 4(1) + (-3) - 2(0) + 2 = 4 - 3 + 0 + 2 = 3$$

$$w_3 = 5(1) - (-3) + 4(0) + 0 = 5 + 3 + 0 + 0 = 8$$

$$T(1, -3, 0, 2) = (1, 3, 8)$$

Compute By Multiplication:-

$$\begin{bmatrix} 2 & -3 & 1 & -5 \\ 4 & 1 & -2 & 1 \\ 5 & -1 & 4 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -3 \\ 0 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 2+9+0-10 \\ 4-3-0+2 \\ 5+3+0+0 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 8 \end{bmatrix}$$

$$T_A(x) = AX.$$

Example : 2 (zero Transformation)

If O is $m \times n$ zero matrix then

$$T_O(x) = OX = O$$

So multiplication by zero maps every vector in R^n to zero vector in R^m . We call T_O the zero Trans.

Example : 3 (Identity operators)

If I is the $n \times n$ identity matrix, then

$$T_I(x) = IX = x$$

So multiplication by I maps every vector in R^n to itself. We call T_I identity operator on R^n .

Properties:

1. $T_A(u+v) = T_A(u) + T_A(v)$ Additive
2. $T_A(kv) = kT_A(v)$ Homogeneity.

Example : 4 (Finding a standard Matrix)

Find Stand Matrix A for Linear Transformation

$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ define by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 + x_2 \\ x_1 - 3x_2 \\ -x_1 + x_2 \end{bmatrix}$$

$$e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$T(e_1) = T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 2(1) + 0 \\ 1 - 3(0) \\ -1 + 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$$

$$T(e_2) = T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 2(0) + 1 \\ 0 - 3(1) \\ -0 + 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \\ 1 \end{bmatrix}$$

$$A = [T(e_1) | T(e_2)] = \begin{bmatrix} 2 & 1 \\ 1 & -3 \\ -1 & 1 \end{bmatrix} \text{ Ans}$$

Example : 5 (computing with stand Matrices)

$T(1, 4)$

$$T\left(\begin{bmatrix} 1 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} 2 & 1 \\ 1 & -3 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 2+4 \\ 1-12 \\ -1+4 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 1 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} 6 \\ -11 \\ 3 \end{bmatrix}$$

Example : 6

Rewrite the transformation $T(x_1, x_2) = (3x_1 + x_2, 2x_1 - 4x_2)$ in vector form and find its standard matrix.

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 3x_1 + x_2 \\ 2x_1 - 4x_2 \end{bmatrix}$$

$$e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$T(e_1) = T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 3(1) + 0 \\ 2(1) - 4(0) \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$T(e_2) = T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 3(0) + 1 \\ 2(0) - 4(1) \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$$

$$A = [T(e_1) \mid T(e_2)] = \begin{bmatrix} 3 & 1 \\ 2 & -4 \end{bmatrix}$$

Exercise : 1.8

Question 1-2

Find Domain and codomain of transformation $T_A(X) = AX$

1(a)-

A has size 3×2

$$\begin{array}{ccc} T_A : \mathbb{R}^2 & \rightarrow & \mathbb{R}^3 \\ \downarrow & & \downarrow \\ \text{Domain} & & \text{Codomain} \end{array}$$

1(b)-

A has size 2×3

$$\begin{array}{ccc} T_A : \mathbb{R}^3 & \rightarrow & \mathbb{R}^2 \\ \downarrow & & \downarrow \\ \text{Domain} & & \text{Codomain} \end{array}$$

1(c)

A has size 3×3

$$\begin{array}{ccc} T_A: \mathbb{R}^3 & \longrightarrow & \mathbb{R}^3 \\ \downarrow & & \downarrow \\ \text{Domain} & & \text{Codomain} \end{array}$$

1(d)

A has size 1×6

$$\begin{array}{ccc} T_A: \mathbb{R}^6 & \longrightarrow & \mathbb{R}^1 \\ \downarrow & & \downarrow \\ \text{Domain} & & \text{Codomain} \end{array}$$

2(a)

A has size 4×5

$$\begin{array}{ccc} T_A: \mathbb{R}^5 & \longrightarrow & \mathbb{R}^4 \\ \downarrow & & \downarrow \\ \text{Domain} & & \text{Codomain} \end{array}$$

2(b)

A has size 5×4

$$\begin{array}{ccc} T_A: \mathbb{R}^4 & \longrightarrow & \mathbb{R}^5 \\ \downarrow & & \downarrow \\ \text{Domain} & & \text{Codomain} \end{array}$$

2(c)

A has size 4×4

$$\begin{array}{ccc} T_A: \mathbb{R}^4 & \longrightarrow & \mathbb{R}^4 \\ \downarrow & & \downarrow \\ \text{Domain} & & \text{Codomain} \end{array}$$

2(d)

A has size 3×1

$$\begin{array}{ccc} T_A: \mathbb{R}^1 & \longrightarrow & \mathbb{R}^3 \\ \downarrow & & \downarrow \\ \text{Domain} & & \text{Codomain} \end{array}$$

Question: 3-4

Find Domain & Codomain of trans. defined by eqs.

3(a)

$$w_1 = 4x_1 + 5x_2$$

$$w_2 = x_1 - 8x_2$$

$$\begin{bmatrix} 4 & 5 \\ 1 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

Domain: $\mathbb{R}^2$ Codomain: $\mathbb{R}^2$

3(b)

$$w_1 = 5x_1 - 7x_2$$

$$w_2 = 6x_1 + x_2$$

$$w_3 = 2x_1 + 3x_2$$

$$\begin{bmatrix} 5 & -7 \\ 6 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$$

Domain: $\mathbb{R}^2$ Codomain: $\mathbb{R}^3$

4(a)

$$w_1 = x_1 - 4x_2 + 8x_3$$

$$w_2 = -x_1 + 4x_2 + 2x_3$$

$$w_3 = -3x_1 + 2x_2 - 5x_3$$

$$\begin{bmatrix} 1 & -4 & 8 \\ -1 & 4 & 2 \\ -3 & 2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$$

Domain: $\mathbb{R}^3$

Codomain: $\mathbb{R}^3$

4(b)

$$w_1 = 2x_1 + 7x_2 - 4x_3$$

$$w_2 = 4x_1 - 3x_2 + 2x_3$$

$$\begin{bmatrix} 2 & 7 & -4 \\ 4 & -3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

Domain: $\mathbb{R}^3$

Codomain: $\mathbb{R}^2$

Question: 5-6

Find domain & codomain of transf defined by matrix product:

5(a)

$$\begin{bmatrix} 3 & 1 & 2 \\ 6 & 7 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Domain: $\mathbb{R}^3$ Codomain = $\mathbb{R}^2$

5(b)-

$$\begin{bmatrix} 2 & -1 \\ 4 & 3 \\ 2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Domain = $\mathbb{R}^2$ codomain = $\mathbb{R}^3$

6(a)

$$\begin{bmatrix} 6 & 3 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Domain = $\mathbb{R}^2$
Codomain = $\mathbb{R}^2$

6(b)

$$\begin{bmatrix} 2 & 1 & -6 \\ 3 & 7 & -4 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Domain = $\mathbb{R}^3$
Codomain = $\mathbb{R}^3$

Question: 7-8

Find Domain & Codomain of Trans. T defined by formula.

7(a)

$$T(x_1, x_2) = (2x_1 - x_2, x_1 + x_2)$$

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 - x_2 \\ x_1 + x_2 \end{bmatrix}$$

Domain = $\mathbb{R}^2$

Codomain = $\mathbb{R}^2$

7(b)

$$T(x_1, x_2, x_3) = (4x_1 + x_2, x_1 + x_2)$$

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} 4x_1 + x_2 \\ x_1 + x_2 \end{bmatrix}$$

Domain = $\mathbb{R}^3$

Codomain = $\mathbb{R}^2$

8(a)

$$T(x_1, x_2, x_3, x_4) = (x_1, x_2)$$

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}\right) = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Domain = $\mathbb{R}^4$

Codomain = $\mathbb{R}^2$

8(b)

$$T(x_1, x_2, x_3) = (x_1, x_2 - x_3, x_2)$$

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 \\ x_2 - x_3 \\ x_2 \end{bmatrix}$$

Domain = $\mathbb{R}^3$

Codomain = $\mathbb{R}^3$

Question 9-10

same as question (7-8)

Question 11-12-

Find stand matrix for trans. defined by eqs.

11 (b)

$$w_1 = 7x_1 + 2x_2 - 8x_3$$

$$w_2 = -x_2 + 5x_3$$

$$w_3 = 4x_1 + 7x_2 - x_3$$

$$A = \begin{bmatrix} 7 & 2 & -8 \\ 0 & -1 & 5 \\ 4 & 7 & -1 \end{bmatrix}$$

part (a) is same.

12 (b)

$$w_1 = x_1$$

$$w_2 = x_1 + x_2$$

$$w_3 = x_1 + x_2 + x_3$$

$$w_4 = x_1 + x_2 + x_3 + x_4$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

part (a) is same.

Question: 13

Find stand matrix for transformation T defined by formula.

a-

$$T(x_1, x_2) = (x_2, -x_1, x_1 + 3x_2, x_1 - x_2)$$

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_2 \\ -x_1 \\ x_1 + 3x_2 \\ x_1 - x_2 \end{bmatrix}$$

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^4$$

$$T(e_1) = T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ -1 \\ 1 + 3(0) \\ 1 - 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 1 \\ 1 \end{bmatrix}$$

$$T(e_2) = T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \\ 0 + 3(1) \\ 0 - 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 3 \\ -1 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \\ 1 & 3 \\ 1 & -1 \end{bmatrix} \quad \text{ANS.}$$

b-

$$T(x_1, x_2, x_3, x_4) = (x_1 + 2x_2 - x_3 + x_4, x_2 + x_3, -x_1)$$

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}\right) = \begin{bmatrix} x_1 + 2x_2 - x_3 + x_4 \\ x_2 + x_3 \\ -x_1 \end{bmatrix}$$

$$T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$$

$$T(e_1) = T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 7(1) + 2(0) - 0 + 0 \\ 0 + 0 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 7 \\ 0 \\ -1 \\ 0 \end{bmatrix}$$

$$T(e_2) = T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 7(0) + 2(1) - 0 + 0 \\ 1 + 0 \\ -0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$T(e_3) = T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 7(0) + 2(0) - 1 + 0 \\ 0 + 1 \\ -0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$T(e_4) = T\left(\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 7(0) + 2(0) - 0 + 1 \\ 0 + 0 \\ -0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 7 & 2 & -1 & 1 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix}$$

C-

$$T(x_1, x_2, x_3, x_4) = (0, 0, 0, 0, 0)$$

zero Transformation.

d-

$$T(x_1, x_2, x_3, x_4) = (x_4, x_1, x_3, x_2, x_1 - x_3)$$

$$T: \mathbb{R}^4 \rightarrow \mathbb{R}^5$$

same procedure will be followed.

Question no: 14-

Find standard matrix for operator T defined by formula.

C-

$$T(x_1, x_2, x_3) = (x_1 + 2x_2 + x_3, x_1 + 5x_2, x_3)$$

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + 2x_2 + x_3 \\ x_1 + 5x_2 \\ x_3 \end{bmatrix}$$

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T(e_1) = T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 + 2(0) + 0 \\ 1 + 5(0) \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$T(e_2) = T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 + 2(1) + 0 \\ 0 + 5(1) \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}$$

$$T(e_3) = T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 + 2(0) + 1 \\ 0 + 5(0) \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

a, b & d : are same.

Question: 15

Find standard matrix for operator T defined by

$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$$w_1 = 3x_1 + 5x_2 - x_3$$

$$w_2 = 4x_1 - x_2 + x_3$$

$$w_3 = 3x_1 + 2x_2 - x_3$$

and compute $T(-1, 2, 4)$ by
directly computing in equations
and then by matrix multiplication

$$A = \begin{bmatrix} 3 & 5 & -1 \\ 4 & -1 & 1 \\ 3 & 2 & -1 \end{bmatrix}$$

Compute by $T(-1, 2, 4)$ direct substituting.

$$w_1 = 3(-1) + 5(2) - 4 = -3 + 10 - 4 = 3$$

$$w_2 = 4(-1) - 2 + 4 = -4 - 2 + 4 = -2$$

$$w_3 = 3(-1) + 2(2) - 4 = -3 + 4 - 4 = -3$$

Compute by Multiplication.

$$\begin{bmatrix} 3 & 5 & -1 \\ 4 & -1 & 1 \\ 3 & 2 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} -3 + 10 - 4 \\ -4 - 2 + 4 \\ -3 + 4 - 4 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ -2 \\ -3 \end{bmatrix}$$

Question: 16

same as q.no 15.

Question: 17-18

Find standard matrix for T and use it to compute $T(x)$. Check your result by subs. directly in formula for T .

17 (a)

$$T(x_1, x_2) = (-x_1 + x_2, x_2); \quad x = (-1, 4)$$

$$Ax = b \rightarrow T(x)$$

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} -x_1 + x_2 \\ x_2 \end{bmatrix} \quad T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(e_1) = T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} -1 + 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$T(e_2) = T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -0 + 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix}$$

Compute $T(x)$

$$\begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 + 4 \\ -0 + 4 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

compute $T(-1, 4)$:

$$T\left(\begin{bmatrix} -1 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} -(-1) + 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

17 (b)

same as part (a)

18 (a)

$$T(x_1, x_2) = (2x_1 - x_2, x_1 + x_2); x = (-2, 2)$$

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 - x_2 \\ x_1 + x_2 \end{bmatrix} \quad T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

SBV:-

$$T(e_1) = T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 2(1) - 0 \\ 1 + 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$T(e_2) = T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 2(0) - 1 \\ 0 + 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$$

Compute $T(x)$

$$\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 - 2 \\ -2 + 2 \end{bmatrix} = \begin{bmatrix} -6 \\ 0 \end{bmatrix}$$

Compute Directly:

$$T\left(\begin{bmatrix} -2 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 2(-2) - 2 \\ -2 + 2 \end{bmatrix} = \begin{bmatrix} -6 \\ 0 \end{bmatrix}$$

18 (b)

Same as part (a).

Question : 19-20

To Find $T_A(x)$ we will perform simple matrix multiplication.

Question : 21-22

Use Theorem 1.8.2 to show that T is a matrix
trans.

$$T(u+v) = T(u) + T(v)$$

$$T(ku) = kT(u)$$

21(a)

$$T(x, y) = (2x+y, x-y)$$

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x+y \\ x-y \end{bmatrix} \quad \text{let } u = \begin{bmatrix} a \\ b \end{bmatrix}, v = \begin{bmatrix} c \\ d \end{bmatrix}$$

$$\text{L.H.S.}:- T(u+v)$$

$$\begin{aligned} T(u+v) &= T\left(\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix}\right) = T\left(\begin{bmatrix} a+c \\ b+d \end{bmatrix}\right) \\ &= \begin{bmatrix} 2(a+c) + b+d \\ a+c - b-d \end{bmatrix} = \begin{bmatrix} 2a+b+2c+d \\ a-b+c-d \end{bmatrix} \end{aligned}$$

$$\text{R.H.S.}:- T(u) + T(v)$$

$$\begin{aligned} T(u) + T(v) &= T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) + T\left(\begin{bmatrix} c \\ d \end{bmatrix}\right) = \begin{bmatrix} 2a+b \\ a-b \end{bmatrix} + \begin{bmatrix} 2c+d \\ c-d \end{bmatrix} \\ &= \begin{bmatrix} 2a+b+2c+d \\ a-b+c-d \end{bmatrix} \end{aligned}$$

Hence $T(u+v) = T(u) + T(v)$ is Linear.

$$T(kU) = kT(U)$$

$$\text{L.H.S. : } T(kU)$$

$$T(kU) = T\left(k \begin{bmatrix} a \\ b \end{bmatrix}\right) = T\left(\begin{bmatrix} ka \\ kb \end{bmatrix}\right) = \begin{bmatrix} 2ka + kb \\ ka - kb \end{bmatrix}$$

$$\text{R.H.S.} = kT(U)$$

$$kT(U) = kT\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = k\left(\begin{bmatrix} 2a+b \\ a-b \end{bmatrix}\right) = \begin{bmatrix} 2ka+kb \\ ka-kb \end{bmatrix}$$

Hence $T(kU) = kT(U)$ is linear trans.

21 (b)-

$$T(x_1, x_2, x_3) = (x_1, x_3, x_1 + x_2)$$

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 \\ x_3 \\ x_1 + x_2 \end{bmatrix}$$

$$\text{L.H.S. : } T(U+V)$$

$$T(U+V) = T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix} + \begin{bmatrix} d \\ e \\ f \end{bmatrix}\right) = T\left(\begin{bmatrix} a+d \\ b+e \\ c+f \end{bmatrix}\right)$$

$$= \begin{bmatrix} a+d \\ c+f \\ a+d+b+e \end{bmatrix}$$

$$\text{R.H.S. : } T(U) + T(V)$$

$$T(U) + T(V) = T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) + T\left(\begin{bmatrix} d \\ e \\ f \end{bmatrix}\right) = \begin{bmatrix} a \\ c \\ a+b \end{bmatrix} + \begin{bmatrix} d \\ f \\ d+e \end{bmatrix}$$

$$= \begin{bmatrix} a+d \\ c+f \\ a+d+b+e \end{bmatrix}$$

Hence $T(U+V) = T(U) + T(V)$ is Linear Trans.

$$T(kU) = kT(U)$$

$$\text{L.H.S.} = T(kU)$$

$$T(kU) = T\left(k \begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = T\left(\begin{bmatrix} ka \\ kb \\ kc \end{bmatrix}\right) = \begin{bmatrix} ka \\ kc \\ ka+kb \end{bmatrix}$$

$$\text{R.H.S.} = kT(U)$$

$$kT(U) = kT\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = k \begin{bmatrix} a \\ b \\ c+a+b \end{bmatrix} = \begin{bmatrix} ka \\ kc \\ ka+kb \end{bmatrix}$$

Hence Homogeneity holds.

22(a)-

$$T(x, y, z) = (x+y, y+z, x)$$
$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$\text{L.H.S.} = T(U+V)$$

$$T(U+V) = T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix} + \begin{bmatrix} d \\ e \\ f \end{bmatrix}\right) = T\left(\begin{bmatrix} a+d \\ b+e \\ c+f \end{bmatrix}\right) = \begin{bmatrix} a+d+b+e \\ b+e+c+f \\ a+d \end{bmatrix}$$

$$\text{R.H.S.} = T(U) + T(V)$$

$$T(U) + T(V) = T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) + T\left(\begin{bmatrix} d \\ e \\ f \end{bmatrix}\right) = \begin{bmatrix} a+b \\ b+c \\ a \end{bmatrix} + \begin{bmatrix} d+e \\ e+f \\ d \end{bmatrix}$$
$$= \begin{bmatrix} a+d+b+e \\ b+e+c+f \\ a+d \end{bmatrix}$$

Additivity holds.

$$T(kU) = kT(U)$$

L.H.S. $T(kU)$

$$T(kU) = T\left(k \begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = T\left(\begin{bmatrix} ka \\ kb \\ kc \end{bmatrix}\right) = \begin{bmatrix} ka+kb \\ kb+kc \\ ka \end{bmatrix}$$

R.H.S. $kT(U)$

$$kT(U) = k T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = k \begin{bmatrix} a+b \\ b+c \\ a \end{bmatrix} = \begin{bmatrix} ka+kb \\ kb+kc \\ ka \end{bmatrix}$$

Homogeneity holds.

22(b)

$$T(x_1, x_2) = (x_2, x_1)$$

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

L.H.S. $T(U+V)$

$$T(U+V) = T\left(\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix}\right) = T\left(\begin{bmatrix} a+c \\ b+d \end{bmatrix}\right) = \begin{bmatrix} b+d \\ a+c \end{bmatrix}$$

R.H.S. $T(U) + T(V)$

$$T(U) + T(V) = T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) + T\left(\begin{bmatrix} c \\ d \end{bmatrix}\right) = \begin{bmatrix} b \\ a \end{bmatrix} + \begin{bmatrix} d \\ c \end{bmatrix}$$

$$= \begin{bmatrix} b+d \\ a+c \end{bmatrix}$$

Additivity holds.

$$T(kU) = kT(U)$$

$$\text{L.H.S: } T(kU)$$

$$T(kU) = T\left(k \begin{bmatrix} a \\ b \end{bmatrix}\right) = T\left(\begin{bmatrix} ka \\ kb \end{bmatrix}\right) = \begin{bmatrix} kb \\ ka \end{bmatrix}$$

$$\text{R.H.S: } kT(U)$$

$$kT(U) = kT\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = k \begin{bmatrix} b \\ a \end{bmatrix} = \begin{bmatrix} kb \\ ka \end{bmatrix}$$

Homogeneity holds.

Question : 23-24 (use Theorem 1.8-2) to Proof T is not a matrix Transform.

23(a)-

$$T(x, y) = (x^2, y)$$

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(U+V) = T\left(\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix}\right) = T\left(\begin{bmatrix} a+c \\ b+d \end{bmatrix}\right) = \begin{bmatrix} (a+c)^2 \\ b+d \end{bmatrix}$$

$$T(U) + T(V) = T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) + T\left(\begin{bmatrix} c \\ d \end{bmatrix}\right) = \begin{bmatrix} a^2 \\ b \end{bmatrix} + \begin{bmatrix} c^2 \\ d \end{bmatrix} = \begin{bmatrix} a^2 + c^2 \\ b+d \end{bmatrix}$$

$T(U+V) \neq T(U) + T(V)$ so not matrix Transform.

$$T(kU) = kT(U)$$

$$T(kU) = T\left(k \begin{bmatrix} a \\ b \end{bmatrix}\right) = T\left(\begin{bmatrix} ka \\ kb \end{bmatrix}\right) = \begin{bmatrix} (ka)^2 \\ kb \end{bmatrix}$$

$$kT(U) = kT\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = k \begin{bmatrix} a^2 \\ b \end{bmatrix} = \begin{bmatrix} ka^2 \\ kb \end{bmatrix}$$

$T(kU) \neq T(U) + T(V)$ so Homog. not Hold.

23 (b)

$$T(x, y, z) = (x, y, xz)$$

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T(U+V) = T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix} + \begin{bmatrix} d \\ e \\ f \end{bmatrix}\right) = T\left(\begin{bmatrix} a+d \\ b+e \\ c+f \end{bmatrix}\right) = \begin{bmatrix} a+d \\ b+e \\ (a+d)(c+f) \end{bmatrix}$$

$$= \begin{bmatrix} a+d \\ b+e \\ ac+af+dc+df \end{bmatrix}$$

$$T(U) + T(V) = T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) + T\left(\begin{bmatrix} d \\ e \\ f \end{bmatrix}\right) = \begin{bmatrix} a \\ b \\ ac \end{bmatrix} + \begin{bmatrix} d \\ e \\ df \end{bmatrix}$$

$$= \begin{bmatrix} a+d \\ b+e \\ ac+df \end{bmatrix}$$

24 (a)

$$T(x, y) = (x, y+1)$$

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(U+V) = T\left(\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix}\right) = T\left(\begin{bmatrix} a+c \\ b+d \end{bmatrix}\right) = \begin{bmatrix} a+c \\ b+d+1 \end{bmatrix}$$

$$T(U) + T(V) = T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) + T\left(\begin{bmatrix} c \\ d \end{bmatrix}\right) = \begin{bmatrix} a \\ b+1 \end{bmatrix} + \begin{bmatrix} c \\ d+1 \end{bmatrix}$$

$$= \begin{bmatrix} a+c \\ b+d+2 \end{bmatrix}$$

$T(U+V) \neq T(U) + T(V)$ so not a matrix trans.

$$T(kU) = kT(U)$$

$$T(kU) = T\left(k \begin{bmatrix} a \\ b \end{bmatrix}\right) = T\left(\begin{bmatrix} ka \\ kb \end{bmatrix}\right) = \begin{bmatrix} ka \\ kb+1 \end{bmatrix}$$

$$kT(U) = kT\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = k\begin{bmatrix} a \\ b+1 \end{bmatrix} = \begin{bmatrix} ka \\ kb+1 \end{bmatrix}$$

Hence $T(kU) \neq kT(U)$ so not matrix trans.
(unless $k=1$)

24(b)

$$T(x_1, x_2, x_3) = (x_1, x_2, \sqrt{x_3})$$

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T(U+V) = T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix} + \begin{bmatrix} d \\ e \\ f \end{bmatrix}\right) = T\left(\begin{bmatrix} a+d \\ b+e \\ c+f \end{bmatrix}\right) = \begin{bmatrix} a+d \\ b+e \\ \sqrt{c+f} \end{bmatrix}$$

$$T(U) + T(V) = T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) + T\left(\begin{bmatrix} d \\ e \\ f \end{bmatrix}\right) = \begin{bmatrix} a \\ b \\ \sqrt{c} \end{bmatrix} + \begin{bmatrix} d \\ e \\ \sqrt{f} \end{bmatrix}$$

$$= \begin{bmatrix} a+d \\ b+e \\ \sqrt{c} + \sqrt{f} \end{bmatrix}$$

$T(U+V) \neq T(U) + T(V)$ coz $(\sqrt{c+f} \neq \sqrt{c} + \sqrt{f})$
so not matrix trans.

$$T(kU) = kT(U)$$

$$T(kU) = T\left(k\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = T\left(\begin{bmatrix} ka \\ kb \\ kc \end{bmatrix}\right) = \begin{bmatrix} ka \\ kb \\ \sqrt{kc} \end{bmatrix}$$

$$kT(U) = kT\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = k\begin{bmatrix} a \\ b \\ \sqrt{c} \end{bmatrix} = \begin{bmatrix} ka \\ kb \\ k\sqrt{c} \end{bmatrix}$$

$T(kU) \neq kT(U)$ coz $(\sqrt{kc} \neq k\sqrt{c})$ so
not a matrix trans.